

“CDs-Maize Power” Utilization of Corn (*Zea mays L.*) Waste C-dots as Environmentally Friendly Batteries

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1. Introduction

1.1 Background

Indonesia is one of the countries with the largest corn production in the world. Based on data from the Food and Agriculture Organization (FAO) published by the Ministry of Agriculture (MOA), Indonesia's average maize production in 2014-2018 was 24.27 million tons. Indonesia's corn production contributed 2.19% to the world's maize production. The 3 regions with the highest maize production in Indonesia are East Java at 5.37 million tons, Central Java at 3.467 million tons, and Lampung at 3.145 million tons. With the large amount of corn production in Indonesia, there is a large amount of corn waste as a by-product of corn itself.

Corn waste is a type of organic waste which contains protein and will produce an unpleasant smell when it decomposes because proteins that contain amine groups will break down into ammonia gas. For human health, the potential health hazards that can be caused are diarrhea and rat diseases due to viruses that come from waste with improper management (Hasibuan R., etc. 2016). In addition, corn waste is still not maximally utilized. Seeing this problem, we need to look for alternative utilization of corn waste so as not to pollute the environment. Corn waste itself can be one of the potential biomass energy sources if managed properly.

In the other hand, the widespread use of lithium batteries, which are included in hazardous waste, has a devastating impact. In the process of mining lithium, it often involves deforestation and water pollution. The lithium battery production process also generates significant carbon emissions. Steps such as raw material extraction, chemical production, and transportation all require fossil energy consumption. Carbon emissions generated by lithium battery production can contribute to climate change and other environmental issues. In addition, lithium batteries require proper waste

management after their end-of-life. Lithium battery waste contains hazardous chemicals that can contaminate soil and water if not managed properly. At the same time, recycling lithium batteries is still a complex and expensive challenge.

Therefore, this research was carried out which can reduce the adverse effects of using waste batteries while utilizing corn waste that is wasted. This paper explores the synthesis of these carbon dots, their properties, and their potential as a sustainable and environmentally friendly alternative to lithium-ion batteries. The research presented in this paper has the potential to contribute to the development of more sustainable and environmentally friendly technologies for the future.

1.2 Objection

1. Does corn waste can be used as an alternative material in the manufacture of environmentally friendly battery?
2. How is the effectiveness comparison between C-dots from Corn Husk, Corn stalks, Corn Leaves, and Corn Cob as an environmentally friendly battery?

1.3 Purpose and Aim

1. To identify the potential of corn waste as an alternative raw material in the manufacture of environmentally friendly batteries.
2. Testing the effectiveness of C-dots from Corn Husk, Corn Stalks, Corn Leaves, and Corn Cob as environmentally friendly batteries.

1.4 Basic Theory

1.4.1 Carbon Nanodots

Carbon dots (C-dots) are new types of fluorescent carbon nanomaterials and are among the most significant members within the carbon nanomaterials family. Generally, C-dots are finely dispersed spherical particles with a size smaller than 10 nm (Kang C., etc. 2020). The controllable

production of carbon dots is currently in its initial stages of development. Synthesis methods can be categorized into two main types: top-down and bottom-up approaches. Bottom-up methods include techniques like hydrothermal, microwave, thermal decomposition, carbonization, pyrolysis, solvothermal, and ultrasonic. Top-down methods include laser ablation, arc discharge, and electrochemical (Khayal A., etc. 2021). The raw materials for creating carbon dots are plentiful and can be categorized into two types: organic and inorganic carbon sources. In practical applications, organic carbon sources such as organic compounds, organic products, and biomass waste are more commonly utilized for synthesizing carbon dots (Cui L., etc. 2021). The use of different synthesis methods and different base materials will result in differences in the levels of carbon, oxygen, and nitrogen formed, surface structure, and functional groups resulting in different characteristics.

1.4.2 Corn Waste

Corn, which has the Latin name *Zea mays var sacchara* Sturt. belongs to the Gramineae family or grasses and is classified as a Monocotyledonous plant. The morphological structure of the corn plant consists of roots, stems, flowers, and fruit (Syukur M.). There are several local Indonesian/regional terms for various kinds of corn plant waste or by-products of corn-based industries such as corn stubble (the entire corn plant including stems, leaves and young corn kernels), corn straw (parts of corn stems and leaves that have been left to dry in the field and harvested when the corn cob is picked), corn husk (the outer skin of corn kernels which is usually discarded), corn cob (waste obtained when corn kernels are threshed from the fruit) (Bunjamin Z., etc. 2014).

1.4.3 AgNO₃

AgNO₃ or silver nitrate compounds are metal compounds derived from silver metal (Ag) reacted with concentrated nitric acid solution (HNO₃ 8 M) which reacts according to the following reaction equation $6\text{Ag (s)} + 8\text{HNO}_3 \text{ (aq)} \rightarrow 6\text{AgNO}_3 \text{ (aq)} + 4\text{H}_2\text{O (l)} + 2\text{NO (g)}$ (Addiin I., etc. 2021). This corrosive compound is arranged in a planar trigonal array and has a molecular weight of 169.873g/mol, Exact Mass of 168.89291 g/mol, Monoisotopic Mass of 168.89291 g/mol, Heavy Atom Count of 5, and Complexity of 18.8 (PubChem. 2021). Silver nitrate remains stable under light exposure and

easily dissolves in water, breaking down into its component ions. It finds use in diverse fields such as photography, the creation of silver-containing explosives, and the manufacture of certain medical substances

1.4.4 Lithium-Ion Battery

Lithium is one of the minerals that has the highest demand in the fourth Industrial Revolution. The main raw material source for batteries is Lithium. (Salafudin. 2021) Lithium is an irreplaceable component in Lithium-ion batteries, and is used as a cathode, electrolyte and anode. Lithium is preferred as a battery material because it is the lightest metal with the highest electrochemical potential among all metals, and has the highest level of energy density among all metals (Bae H.,etc.2021)

1.4.5 Swagelok

The swagelok type cell battery fixture is used for lithium batteries, Zinc ion batteries, sodium ion batteries and solid battery research work, it can replace of coin cell cases to save the cost. A swagelok is a simple cylindrical electrochemical cell consisting of two electrodes, a separator, and a Swagelok fitting, which is a stainless-steel compression fitting that holds the cell assembly together and provides an electrical connection to the electrodes. The Swagelok cell is a custom electrochemical cell that allows for easy assembly and disassembly in an airtight system. Swagelok type cell is specially designed for lithium-ion battery's research. Cells are useful for studying the behavior of single or full electrodes using foil or carbon cloth electrodes as anode and cathode electrodes, this is to achieve super capacitors or Li-ion batteries that are close to the device system with very minimal steps and can perform various types of measurements such as CV, EIS, and Charge-discharge.

2. Method and Experimental Details

2.1 Time and Place

This research was conducted in July 2024 at researcher's house and STIFAR Laboratory.

Date	Activity	Activity Details	Place
30 June – 10 July 2024	Planning	Literature studies about Corn, Carbon Nanodots, Battery cells, AgNO ₃ , Swagelok, and FTIR Method	Researcher's house
11 – 13 July 2024	Research Design	Looking for tools and material for research proses	Lab store & Corn fields
14 – 21 July 2024	Research Process	Making C-dots, FTIR Test, AgNO ₃ synthesis, Making AgNO ₃ C-dots Battery Tablet	Researcher's house & STIFAR Laboratory
22 – 31 July 2024	Analyze	Discussion to analyze and write the research result	Researcher's house

2.2 Research Design



2.3 Experiment Method

2.3.1 Tools and Material

Material needed for research are:

1. Dry Corn Husk
2. Dry Corn Stalks
3. Dry Corn Leaves
4. Dry Corn Cob
5. Aquadest
6. AgNO_3 crystals
7. KOH
8. Ethylene Glycol
9. 70% Alcohol
10. Paper filters

Tools needed for research are:

1. Microwave
2. Beaker glass
3. Watch glass
4. Porcelain mortars
5. Mg/Micro unit scales
6. Swagelok
7. Multimeter
8. Laboratory water bath
9. Blender
10. Knife

2.3.2 Experiment Process

2.3.2.1 Making Carbon Dots

1. Prepare all tools and material
2. Cut material into small pieces



3. Blend with blender to create smaller particle



4. Microwave at 450 watts for 20 minutes x 5



5. Filter Carbon Dots with aquadest



6. Dry Carbon Dots for 6 hour



7. Grind Carbon Dots into fine particles using mortar



8. Carbon dots are ready to be used

2.3.2.2 Synthesis of C-dots with AgNO_3

1. 10g of AgNO_3 is added into 10g of C-dots that has been dispersed into 1000ml of 70% alcohol.



2. 300ml of ethylene glycol and 2g of KOH is also added. Then, reduced in 85°C water bath while being stirred constantly for 60 minutes.



3. Separate the sediment from the solution.
4. Wash the sediment with aquadest and dried at 50°C overnight in an oven.
5. Reduced C-dots without AgNO_3 is also prepared as comparison

2.3.2.2 Making & Testing CDs-Maize Power Tablet

1. Insert 1g of C-dots into Swagelok (pressing tools)
2. The process of pressing is done manually until a tablet is formed.
3. Reduced C-dots without AgNO₃ is also prepared for independent study as a comparison.
4. Test and compare the result CDs-Maize Power millivolt and microampere using multi meter.

3. Result and Discussion

3.1 Multi-meter Test

a. C-dots

No	C-dots	Millivolt (mV)	Milliampere (mA)
1	Corn Husk	1.4	0
2	Corn Stalks	0.3	2
3	Corn Leaves	10.6	0
4	Corn Cob	0.4	0

b. C-dots/Ag

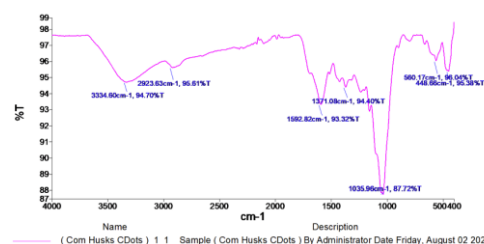
No	C-dots/Ag	Millivolt (mV)	Milliampere (mA)
1	Corn Husk	51.6	8
2	Corn Stalks	5.4	5.2
3	Corn Leaves	149.6	2.9
4	Corn Cob	11.4	3.8

Based on the result of Multimeter test on 4 C-dots/Ag which are Corn Husk, Corn Stalk, Corn Leave, and Corn Cob, it can be seen that Corn Leave C-dots produces the highest voltage with 149.6mV and 2.9mA compared to Corn Husk C-dots/Ag with 51.6 mV and 8mA, Corn Stalks C-dots/Ag with 5.4mV and 5.2mA, and Corn Cob C-dots/Ag with 11.4mV and 3.8mA. It can be concluded that we can utilize C-dots/Ag from Corn Leave as an ingredient to construct an alternative to lithium-ion batteries

3.2 FTIR Test

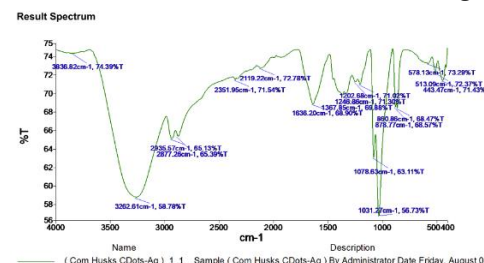
a. Corn Husk C-dots

FTIR Result of Corn Husk C-dots



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3334.60	94.70
C-H	2923.63	95.61
N=H	1592.82	93.32
C=C	1371.08	94.40
C-C	1035.95	87.72

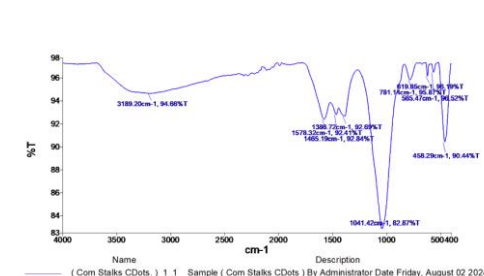
FTIR Result of Corn Husk C-dots/Ag



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3334.60	94.70
C-H	2923.63	95.61
N=H	1592.82	93.32
C=C	1371.08	94.40
C-C	1035.95	87.72

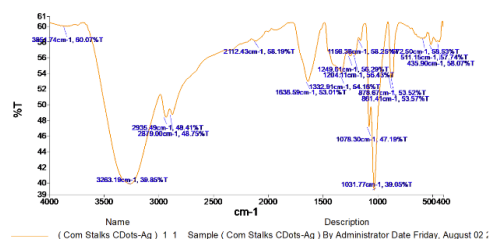
b. Corn Stalks C-dots

FTIR Result of Corn Stalks C-dots



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3334.60	94.70
C-H	2923.63	95.61
N=H	1592.82	93.32
C=C	1371.08	94.40
C-C	1035.95	87.72

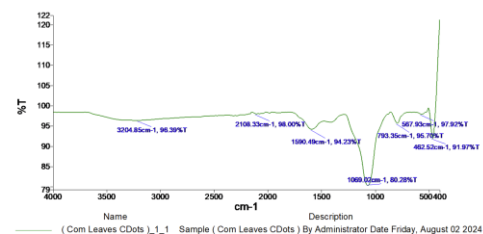
FTIR Result of Corn Stalks C-dots/Ag



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3263.19	39,85
C-H	2879.00	48,75
N=H	1636.59	53,01
C=C	1332.91	54,16
C-C	1031.77	39,05

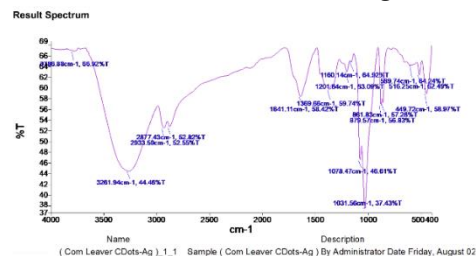
c. Corn Leaves C-dots

FTIR Result of Corn Leaves C-dots



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
N=H	1590.49	94.23
C-C	1069.02	80.28

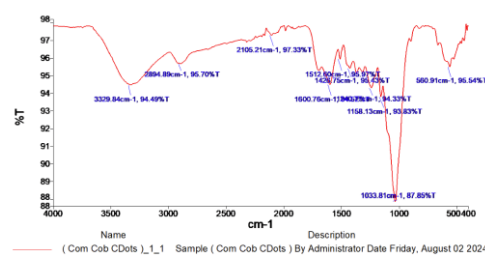
FTIR Result of Leave C-dots/Ag



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3261.94	44,46
C-H	2933.59	52,55
N=H	1641.11	58,45
C=C	1369.66	59,74
C-C	1031.56	37,43

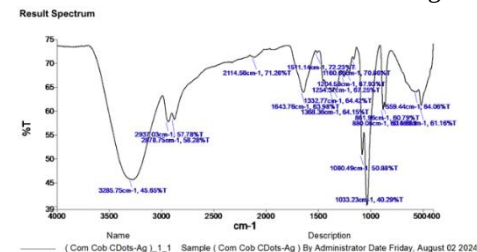
d. Corn Cob C-dots

FTIR Result of Corn Cob C-dots



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3329.84	94.49
C-H	2894.89	95.70
N=H	1600.76	94.52
C=C	1426.75	95.43
C-C	1033.81	87.85

FTIR Result of Corn Cob C-dots/Ag



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3285.75	45,65
C-H	2937.03	57,78
N=H	1643.76	63,98
C=C	1332.77	64,42
C-C	1033.23	40,29

The results of the FTIR testing of C-dots are presented in the figure and table above. Based on the FTIR analysis of C-dots before and after the AgNO₃ synthesis treatment, differences in functional group amount were observed. In Corn Husk C-dots, the OH bond was detected at a wavenumber of 3334.60 cm⁻¹ with a transmittance of 94.70%, while the C-H bond was observed at a wavenumber of 2923.63 cm⁻¹ with a transmittance of 95.61%. Conversely, in Corn Husk C-dots/Ag, the OH bond was identified at a wavenumber of 3262.61 cm⁻¹ with a transmittance of 58.78%, and the C-H bond was detected at a wavenumber of 2877.26 cm⁻¹ with a transmittance of 65.39%.

This also applies to other C-dots. In Corn Stalks C-dots, the OH bond was detected at a wavelength of 3189.20 cm⁻¹ with a transmittance of 94.66%, whereas the C-H bond was not detected. In Corn Stalks C-dots/Ag, the OH bond was identified at a wavelength of 3263.19 cm⁻¹ with a transmittance of 39.85%, while the C-H bond

was also observed.

The FTIR results for Corn Leaves C-dots indicated that neither the OH nor C-H bonds were detected. However, in Corn Leaves C-dots/Ag, the OH bond was detected at a wavelength of 3261.94 cm^{-1} with a transmittance of 44.46%, and the C-H bond was observed at a wavelength of 2933.59 cm^{-1} with a transmittance of 52.55%.

FTIR results for Corn Cob C-dots, OH bond was detected at a wavelength of 3329.84 cm^{-1} with a transmittance of 94.49%, and the C-H bond was detected at a wavelength of 2894.89 cm^{-1} with a transmittance of 95.70%. On the other hand, in Corn Cob C-dots/Ag, the OH bond was identified at a wavelength of 3285.75 cm^{-1} with a transmittance of 45.65%, as well as the C-H bond at a wavelength of 2937.03 cm^{-1} with a transmittance of 57.78%.

The FTIR results of C-dots obtained from corn waste showed significant modifications in functional groups before and after AgNO_3 treatment, which has important implications in their application as battery anodes. In C-dots from corn husk, there was a decrease in transmittance for OH groups from 94.70% to 58.78% and for C-H from 95.61% to 65.39%. This decrease indicates that the interaction between silver ions and hydroxyl groups occurs, which can improve the ionic conductivity ability and stability of the electrode. C-dots from stalk and cob also showed an addition of C-H groups after treatment, signifying an increase in energy storage potential through better redox reactions, which is crucial for improving battery charge and discharge efficiency.

In contrast, in C-dots from leaves, which previously showed no OH or C-H groups, AgNO_3 treatment successfully produced both groups, with a fairly low transmittance. This indicates that the synthesis process not only introduced new groups but also increased the potential of this material for battery applications. Overall, the presence of still detectable hydroxyl groups and the addition of C-H groups in most of the sources indicate that corn waste C-dots, after modification, have good potential as battery anodes, given the importance of electrochemical properties in efficient energy storage applications.

4. Conclusion

It can be concluded that Corn Leave C-dots/Ag has the best potential as an ingredient to construct an alternative to lithium-ion batteries.

FTIR results showed that the synthesis of AgNO_3 on C-dots from corn waste caused significant changes in functional groups, which correlated with an increase in voltage of 139.0 mV and 2.9 mA on corn leaves. In corn leaves, before AgNO_3 treatment, no OH or C-H bonds were detected. After treatment, an OH bond appeared at 3261.94 cm^{-1} with a transmittance of 44.46% and a C-H bond at 2933.59 cm^{-1} with a transmittance of 52.55%. The appearance of these OH and C-H groups indicates an increase in carbon content and ion transportation ability, which can improve conductivity and electrochemical efficiency. This explains why C-dots from corn leaves produce the highest voltages compared to other corn parts, making them excellent candidates for battery anode materials.

5. Acknowledgements

Finally, our project is finished. We would like to express our deepest gratitude to Drs. Agus Priyatno, M.Pd., for his invaluable guidance, support, and encouragement throughout the course of this research. His expertise and insights have been instrumental in the successful completion of this study. We also extend our appreciation to all those who have contributed to this project, directly or indirectly, for their assistance and cooperation.

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